

ECON 311

Lecture 10: Long-Run Growth

Economics of ideas

Growth Accounting

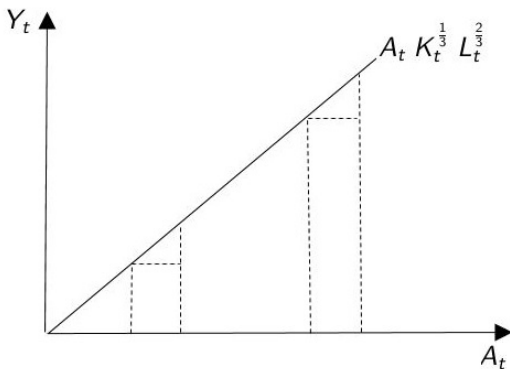
Inflation

Ideas in the Production Function

- Production of ideas result in changes in TFP:

$$Y_t = F(K_t, L_t, A_t) = A_t K_t^{\frac{1}{3}} L_t^{\frac{2}{3}}$$

- There are constant returns to TFP



Ideas in the Production Function

- Production of ideas result in changes in TFP:

$$Y_t = F(K_t, L_t, A_t) = A_t K_t^{\frac{1}{3}} L_t^{\frac{2}{3}}$$

- There are increasing returns to scale

- ▶ Multiply all inputs by a constant (γ):

$$Y_t = F(\gamma K_t, \gamma L_t, \gamma A_t) = \gamma A_t (\gamma K_t)^{\frac{1}{3}} (\gamma L_t)^{\frac{2}{3}}$$

$$Y_t = \gamma^2 F(K_t, L_t, A_t)$$

The Production of Ideas

- Producing ideas is not free:
 - ▶ Innovators must invest “labour” to produce ideas:

$$L_{A_t} = \bar{l} \bar{L}$$

$$L_{Y_t} = (1 - \bar{l}) \bar{L}$$

$$L_{A_t} + L_{Y_t} = \bar{L}$$

- ▶ Trade-off between producing consumption goods and ideas:

$$Y_t = A_t L_{Y_t}$$

$$A_{t+1} = A_t + \underbrace{\bar{z} A_t L_{A_t}}_{\text{Idea}}$$

TFP Growth

- Accumulation of ideas result in TFP growth over the long-run:

$$\underbrace{\frac{A_{t+1} - A_t}{A_t}}_{\bar{g}} = \bar{z} \bar{l} \bar{L}$$

- (Trade-off) An increase in $\bar{l}$ increases A_{t+1} , but decreases Y_t
 - ▶ An increase in $\bar{l}$ has the potential to increase output in the long-run
 - ▶ Furthermore, an increase in $\bar{l}$ increases the long-run growth rate of output

Output Growth in the Long-Run

- Output per capita at time t is:

$$y_t = A_0 (1 - \bar{l}) (1 + \bar{g})^t$$

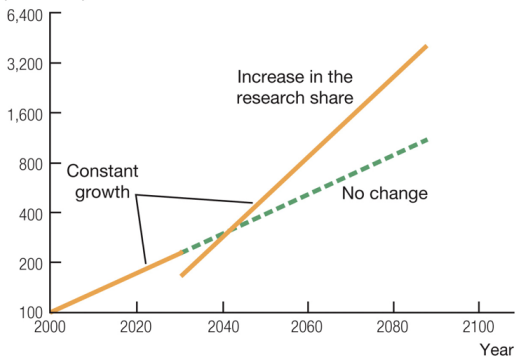
- Using the growth rate of TFP, we can solve for the long-run growth of y_t (BGP):

$$g_y = \bar{z} \bar{l} \bar{L}$$

Trade-Off

Output per Person after an Increase in $\bar{\ell}$

Output per person, y_t
(ratio scale)



Conclusion

- Introducing the production of ideas ...
 - ▶ provides a theory as to why, in the long-run, some countries are able to grow at a faster rate than others
 - ▶ consistent with the data models long-run growth in GDP per capita as the result of TFP growth
 - ▶ provides an explanation as to why some countries have a higher level of TFP growth (better ideas generation process)

Economics of ideas

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Solow and Romer

We combine Romer and Solow:

$$Y_t = F(K_t, L_t, A_t) = A_t K_t^{\frac{1}{3}} L_t^{\frac{2}{3}}$$

$$L_{A_t} = \bar{l} \bar{L}$$

$$L_{Y_t} = (1 - \bar{l}) \bar{L}$$

$$\underbrace{\frac{A_{t+1} - A_t}{A_t}}_{g_{A_t}} = \bar{z} L_{A_t}$$

$$\underbrace{\frac{K_{t+1} - K_t}{K_t}}_{g_{K_t}} = \bar{s} \frac{Y_t}{K_t} - \bar{d}$$

Solow and Romer

- In the long-run equilibrium (BGP), we need capital to grow at a constant rate, g_K , which means:

$$g_Y = g_K$$

- Using the above equation and the growth rate of TFP, we can solve for the long-run growth of Y_t :

$$g_Y = g_A + \frac{1}{3} g_k + \frac{2}{3} g_{(1-\bar{l})\bar{L}}$$

$$g_Y = \bar{z}\bar{l}\bar{L} + \frac{1}{3} g_y + \frac{2}{3} 0$$

$$g_Y = g_y = \frac{3}{2} \bar{z}\bar{l}\bar{L} \geq 0$$

Growth Accounting

- Similar to the Production Function analysis, Growth Accounting disaggregates growth of output into the growth of inputs:

$$g_y = g_A + \frac{1}{3}(g_k - g_L) + \frac{2}{3}(g_{L_{Y_t}} - g_{L_t})$$

Growth Accounting for the United States

	1948–2014	1948–1973	1973–1995	1995–2007	2007–2014
Output per hour, Y/L	2.4	3.3	1.6	2.8	1.4
Contribution of K/L	0.9	1.0	0.8	1.1	0.6
Contribution of labor composition	0.2	0.2	0.2	0.2	0.3
Contribution of TFP, A	1.3	2.1	0.6	1.5	0.5

The table shows the average annual growth rate (in percent) for different variables.
Source: Bureau of Labor Statistics, *Multifactor Productivity Trends*.

Economics of ideas

Growth Accounting

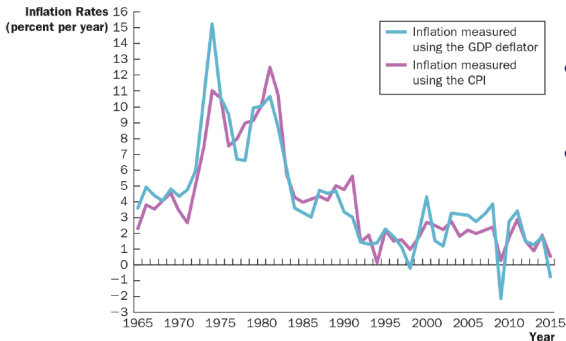
Inflation

Introduction

- Inflation rate is defined as the rate of change of prices

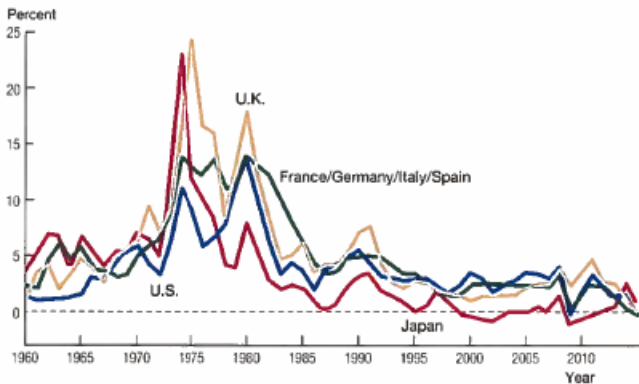
$$\pi_{t+1} = \frac{P_{t+1} - P_t}{P_t}$$

- Inflation in the Canadian economy:

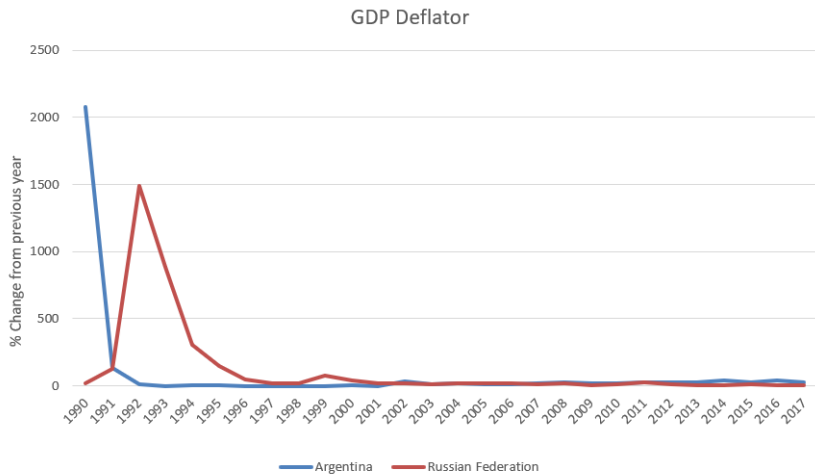


Introduction

Inflation Rates in Certain Rich Countries



Introduction



The Quantity Theory of Money

- The central component of the quantity theory of money is the quantity equation, which connects nominal GDP with the effective amount of liquidity in circulation:

$$M_t V_t = P_t Y_t$$

- ▶ M_t =Money Supply
 - ▶ V_t =Velocity (the rate at which money changes hands)
 - ▶ P_t =Price level
 - ▶ Y_t =Real GDP
- Effective money supply is equal to money demand

Prices and Inflation in the Quantity Theory

- Classical Dichotomy: real and nominal sides of the economy are separate (money is **neutral** in the long run).
- Prices in the quantity theory of money are given by:

$$P_t = \frac{M_t \bar{V}}{Y_t}$$

- ▶ Prices are given by how many goods the effective supply of money is chasing.
- Inflation is given by the following equation:

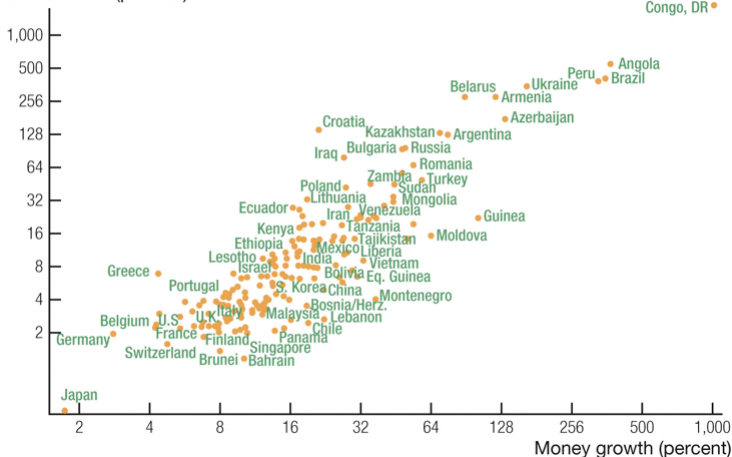
$$\Pi = g_M - g_Y$$

- ▶ If the classical dichotomy holds, we should observe that, on average and over the long-run, high inflation correlates with high levels of money supply growth.

Inflation in the Quantity Theory

Money Growth and Inflation around the World, 1990–2011

Inflation rate (percent)



Risk of Inflation

- As we will see in the short-run analysis, monetary policy is an important tool to fight recessions (such as the COVID recession)
- A risk with loose monetary policy is the risk of increasing the inflation rate over the long-run
 - ▶ Costs associated with large levels of inflation are well understood.
 - ▶ Large debate over whether this risk has been overblown by economists in recent times (MMT)